

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
 Department of Computer Science & Engineering
MID SEMESTER EXAMINATION, March. 2024
B. Tech: CSE, Semester-VI

Course Name: Network Security

Code: CS64153

Maximum Time: 2 hours

Max. Marks: 30

Instruction:

1. Attempt all questions.
2. Assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand margin.

1.	a. Define Internet key exchange (IKE) and explain why it is needed in IPSec. b. A host receives an authenticated packet with the sequence number 331. The replay window spans from 200 to 263. What will the host do with the packet? Explain briefly. What is the window span after this event?	3+3	CO1, CO2	U, A
2.	A stream cipher which has two phases a key generation phase and an encryption phase. a. Explain a stream cipher which is used in GSM network for secure data transmission. b. Given the superincreasing tuple $b = \{ 7, 11, 23, 43, 87, 173, 357 \}$, $r=41$, modulus $n= 1001$, encrypt and decrypt "a" using the knapsack cryptosystem. Use [7 6 5 1 2 3 4] as the permutation table.	3+3	CO2	E
3.	a. Describe how key materials are created from master secret in SSL. b. Show how SSL or TLS reacts to a replay attack. That is show how SSL or TLS responds to an attacker that tries to replay one or more handshake messages.	3+3	CO3	A, U
4.	a. What are the requirements for Kerberos in an open distributed environment? b. Explain the message exchange to obtain Ticket grant, Service grant and target service.	3+3	CO2	U, R
5.	a. In PGP explain how Bob and Alice exchange the secret key for encrypting message. b. Mention the three ways of Denial of Service (DoS) attack which can be happened in a network.	3+3	CO4	U,P

End of Questions